

Yiming (Emmett) Peng

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EDUCATION

- **University of Toronto** Sept 2026 – May 2030 (Expected)
Doctor of Philosophy in Biostatistics
Research interests: causal inference and machine learning
◦ Advisors: Kuan Liu, Zihang Lu
- **University of Toronto** Sept 2024 – Aug 2026
Master of Science in Biostatistics – Thesis Option; GPA: 4.00/4.00
Thesis: Bayesian Latent Class Causal Models for Multiple and Longitudinal Cognitive Outcomes in Aging
◦ Advisors: Kuan Liu, Zihang Lu
- **Beijing Institute of Technology, Zhuhai** Sept 2020 – Jun 2024
Bachelor of Science in Applied Statistics; GPA: 3.96/4.00 (WES)
Thesis: Deep Learning and Stochastic Modeling of Human Activity Patterns through Smartwatch Sensors for Health Interventions (*Honored as Outstanding Thesis*)
◦ Advisor: Han C.W. Hsiao

PUBLICATIONS AND MANUSCRIPTS

P=PUBLISHED, R=UNDER REVIEW, S=IN SUBMISSION, W=WORKING PAPERS

- [P] Denier, D., Peng, Y. E., . . . , Ricciuto, A. (2026). **Longitudinal GGT Trajectories Identify Prognostic Phenotypes in Paediatric Primary Sclerosing Cholangitis**. *JHEP Reports*, 8(10), 101940. [doi:10.1016/j.jhepr.2026.101940](https://doi.org/10.1016/j.jhepr.2026.101940)
- [R] Peng, Y. E.*, Truong, V.*, Xie, A., Shi, Y., Hu, P. (2026). **PRECISE: A Framework for Predicting Treatment Response Using Cell-Type Inference and Single-Cell Embeddings in Breast Cancer Patients Receiving Anti-PD-1 Treatment**. *Under Review*. [\[Code\]](#)
- [R] Shi, Y.*, Zhang, J.*, Xie, A.***, Zhang, J. Z.***, . . . , Peng, Y. E., . . . , Hu, P. (2026). **CRISP-AF: Competing-Risk Interpretable Survival Prediction for Atrial Fibrillation Using an Uncertainty-Calibrated Transformer in 106,651 CIROC Participants**. *Under Review*. [\[Code\]](#)
- [S] Peng, Y. E., Lu, Z., Anderson, G. M., Liu, K. (2026). **Bayesian Latent Class Approach for Causal Estimation with Multiple Outcomes**. *In Submission*.
- [W] Peng, Y. E., Lu, Z., Liu, K. (2026). **Bayesian Nonparametric Latent Class Models for Causal Inference in Cognitive Trajectories**. *Manuscript in Preparation*.
- [W] Jing, Y., Peng, Y. E., Danguedan, A., Knight, A. (2026). **Examining the Relationship Between Socioeconomic Factors and Mental Health in Childhood-Onset Systemic Lupus Erythematosus Using a Multilevel Analysis**. *Manuscript in Preparation*.

PRESENTATIONS

- [1] Peng, Y. (Emmett), Lu, Z., Liu, K. (2026). **Bayesian Nonparametric Latent Class Framework for Causal Inference in Cognitive Trajectories**. Oral presentations at the *Statistical Society of Canada (SSC) Annual Meeting and Joint Statistical Meetings (JSM)*, and a poster presentation at the *ICSA–Canada Chapter Symposium*, 2026. [\[Slides\]](#) [\[Poster\]](#)
- [2] Peng, Y. (Emmett), Denier, D., . . . , Ricciuto, A. (2026). **Longitudinal GGT Trajectories Identify Prognostic Phenotypes in Paediatric Primary Sclerosing Cholangitis**. Poster presentation, *Canadian Statistics Student Conference (CSSC)*, Hamilton, Ontario, Canada, May 2026. [\[Poster\]](#)
- [3] Peng, Y. (Emmett), Truong, V., Xie, A., . . . , Hu, P. (2025). **Enhancing Breast Cancer Treatment Response Prediction with Single-Cell RNA Sequencing and Large Language Models**. Oral presentation and poster, *2025 Statistical Society of Canada (SSC) Annual Meeting*, p. 58. May 25–28, Saskatoon, Saskatchewan, Canada. [\[Slides\]](#) [\[Poster\]](#)
- [4] Zhang, J., Shi, Y., Xie, A., Peng, Y. (Emmett), Hu, P. (2025). **Uncertainty-Calibrated Interpretable Tabular Transformer Model for Atrial Fibrillation Prediction with Competing Risk**. Poster, *2025 Statistical Society of Canada (SSC) Annual Meeting*, p. 42. May 25–28, Saskatoon, Saskatchewan, Canada. [\[Poster\]](#)
- [5] Peng, Y. (Emmett), Lu, Z., Liu, K. (2025). **Bayesian Latent Class Causal Inference with Nonparametric Covariate Modeling for Endpoint Cognitive Outcomes in Aging Populations**. Invited poster presentation, *Data Science Institute Causal Inference Workshop*, Toronto, Ontario, Canada, October 22, 2025. [\[Poster\]](#)

PROFESSIONAL EXPERIENCE

- Knight Lab, The Hospital for Sick Children** Jun 2025 - Present
Clinical Research Project Assistant, supervised by Dr. Andrea Knight
Toronto, Canada
 - Conduct statistical analyses in R for lupus studies, including Bayesian modeling, latent-class trajectory analysis, and simulation-based power analysis for study design and grant applications.
- The Hospital for Sick Children** Jun 2025 - Present
Research Volunteer, supervised by Dr. Amanda Ricciuto and Dr. Anne Griffiths
Toronto, Canada
 - Applied latent class mixed models and survival analysis in R to characterize longitudinal biomarker trajectories in PSC and assess their association with transplant outcomes.
- Hu Lab, Dalla Lana School of Public Health, University of Toronto** Oct 2024 - Aug 2025
Data Science Research Student, supervised by Dr. Pingzhao Hu
Toronto, Canada
 - Developed a Python-based prediction framework integrating single-cell RNA-seq, cell-type inference, and embeddings from language and foundation models to predict breast cancer treatment response.
- The University of Hong Kong** Jan 2025 - Apr 2025
Research Volunteer, supervised by Dr. Liwu Zheng
 - Provided statistical analysis and consulting support for clinician-led biomedical research projects.

TEACHING EXPERIENCE

- STA314H1: Statistical Methods for Machine Learning I** Fall 2026
Teaching Assistant
- STAB22: Statistics I** Summer 2026
Teaching Assistant

HONORS AND AWARDS

Amounts in Canadian dollars unless otherwise specified

- PHS Student Conference Award - \$150** 2026
University of Toronto
- ICSA-Canada Chapter Symposium Student/Postdoc Travel Scholarship Award - \$600** 2026
ICSA-Canada Chapter Symposium
- Winner – Case Studies in Data Analysis Competition, SSC 2025 - \$2,500 per team** 2025
Statistical Society of Canada (SSC)
- CSSC 2025 Travel Award - \$150** 2025
Canadian Statistics Student Conference (CSSC), sponsored by CANSSI
- SGS Conference Grant - \$450** 2025
University of Toronto
- BITZH President’s Scholarship - \$2,000** 2023
Beijing Institute of Technology, Zhuhai
- College Dean’s Honor Scholarship - \$7,000** 2021 - 2024
Beijing Institute of Technology, Zhuhai
- BITZH Freshman Entrance Scholarship and Continuing Scholarship - \$18,000** 2020 - 2024
Beijing Institute of Technology, Zhuhai
- BITZH Outstanding Student Scholarship** 2020 - 2024
Beijing Institute of Technology, Zhuhai

SKILLS

- Programming Languages:** R, Python, MATLAB, C++
- Tools & Technologies:** Git, Linux, HPC (Compute Canada, Slurm), LaTeX, RMarkdown, Jupyter Notebook
- Statistical Methods:** Causal Inference, Latent Class Analysis, Survival Analysis, Longitudinal Data Analysis
- Machine Learning Models:** Transformers, RNNs, Regression (GLM, Penalization), CNNs, Neural Networks
- Languages:** Chinese (native), English (professional working proficiency)

COMMUNITY INVOLVEMENT AND LEADERSHIP EXPERIENCE

- ASA DataFest at UofT** May 2026
Judge and Mentor 
- Biostatistics Union of Graduate Students (BUGS)** Sep 2024 – Present
Seminar Committee member 
- Health Data Working Group** Sep 2024 – Present
Student Member 
- Public Health Students’ Association, University of Toronto** Sep 2025 - Jun 2026
Biostatistics MSc Representative
- Student Union, College of Global Talent, Beijing Institute of Technology, Zhuhai** Sep 2020 – Jun 2022
Officer